

Q1. Given a sorted array containing N integers, both positive and negative. Create another array containing the square off all the elements in the input and return in non-decreasing array.

INPUT : -7 -5 -2 0 1
OUTUT : 0 1 4 25 49

Q2. Provided a sorted linked list, write a program to delete all the nodes that have duplicate numbers, leaving the linked list having unique numbers.

INPUT : 2->4->5->5->6->6->8
OUTPUT : 2->4->5->6->8

Q3. Given a binary tree, print the node values in any order which follow special sum property. Special sum property is having sum of all the left child's only, equal to the sum of all right child's only.

Input : 26 , 18 , 30 , 10 , 21 , 7 , 5 , 8 , 11 , n , n , n , n , 12 , 2 , 3 , 4 , 6 , n , n , n , n , n (level order)
output : 10 , 18 , 30

Q4. Given a valid expression in linked list, print the number of redundant parentheses.

Note– Don't reduce (a+b) to a+b

I/P – ((a-b)+(((c+d)))+(e*f))

O/P – 2

Explanation– (((c+d))) can be reduced to (c+d)

Q5. Given a binary search tree and a range [L, R], delete all elements which are not in the range.

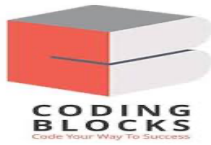
Q6. Given a array of size N and a element K, we have to rearrange the element in such a way that all the element values less than K are at beginning (in any order) followed by all the elements values greater than K (in any order) and all the elements having equal value to k at the end/

INPUT : [6,7,2,5,4,9,8,5], K= 5
OUTPUT : [2,4,6,7,9,8,5,5]

Q7. A string contains only two characters 'I' and 'D' where 'I' represents the increasing sequence and 'D' represents the decreasing sequence. Find the minimum number without repetition of the numbers such that it follows the sequence given in the string. The numbers used in the sequence must be greater than zero. We have to print 'String not found' if the size of the string is zero and if the size of the string is greater than 8 we have to print 'String length exceeds'.

INPUT: IDIDI
OUTPUT: 132546





Q8. We are given a binary tree. The task is to transform every left child node odd by subtracting one and even child node even by adding one to it. We have to write solutions in linear time complexity and constant space.

We are given an array of sizes 'N'. The task is to count greater elements on the left side of each array element.

INPUT: 5

1 2 3 4 5

OUTPUT: -1 1 2 3 4

Q9. You are given an array `arr[]` of N integers including 0. The task is to find the smallest positive number missing from the array.

Q10. Given a binary tree. Find the size of its largest subtree which is a Binary Search Tree

Q11. (1) flattening linked list (2) find the maximum sum in Binary Tree without including adjacent element

Q12. Write a SQL query to get some data from two tables with a condition given?

Q13. Write a code for checking a char array is a palindrome or not?

Q14. Rain water Trapping(Leetcode). 2) Rotate the for the k places.

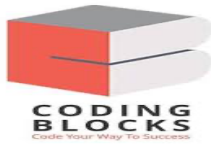


Q15 Write a program to add two Linked Lists.

Q16 . Rotate binary tree (print it as if it were rotated by 90 degrees, don't actually rotate or change the tree)

Q17. Flatten a linked list
Reorder linked list
2-D view of binary tree
Largest BST in binary tree

Q18. One question was based on Zig-Zag traversal in tree.
Other was finding the sum of nodes which are at distance k from the root node.



Q19. Given a linked list, arrange the linked list in the manner of alternate last and first. Make sure to update the node's position instead of just updating the values.

Q20. Flatten a binary tree into linked list
Wave sort a linked list

Q21. Kruskal's Algorithm

Q22. check for balanced parenthesis in an expression

Q23. Write a function to find the intersection point of two singly linked list

Q24. 0/1 knapsack problem

Q25 . Find the longest palindromic substring in a given string

Q26. Find the shortest path in an unweighted graph using BFS



**CODING
BLOCKS**
Code Your Way To Success